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Dear Editors at *Methods in Ecology and Evolution*,

We are submitting a manuscript, entitled *The Bayesian Inference library for Python R and Julia* for consideration as a *Resources* in *Methods in Ecology and Evolution*. Our work introduces a comprehensive suite of novel statistical software for Bayesian analysis that ensures native interoperability across Python, R, and Julia. The framework is designed to bridge the gap between novice and expert users by offering variable levels of abstraction, while simultaneously leveraging hardware acceleration to achieve high scalability and rapid computation.

Bayesian modeling has emerged as an essential part of the toolbox of modern statistics and machine learning, providing a framework for robust inference under uncertainty. But the current ecosystem of Bayesian software is difficult for many end-users (i.e., academic researchers) to navigate. Key challenges stem from the diverse array of different programming languages and packages users may be required to learn, trade-offs between ease-of-use and flexibility, and persistent computational scalability limitations for complex models or large datasets.

We introduce *Bayesian Inference (BI)*, a software package designed to unify the modeling experience across the three dominant data-science languages, Python, R and Julia. *BI* tackles the *interoperability* barrier head-on by offering native interfaces in all environments. It aims to resolve the *accessibility-flexibility trade-off* by providing an intuitive model definition syntax, while permitting advanced customization. *BI* also includes pre-built functions tailored for specialized models in areas like network analysis, survival analysis, and phylogenetic analysis, while still allowing extension and modification within its general framework. Crucially, *BI* enhances *scalability* by integrating with hardware-accelerated computation via *JAX*, which lets users choose between execution on CPUs, GPUs, and TPUs.

BI is designed to be modular and extensible, allowing users to easily incorporate their own custom models and data types into the framework. One of the key features of this software is its comprehensive library of 27 predefined Bayesian models available in the official website, covering a wide range of common applications and use cases. These models are accompanied by detailed explanations, making it easier for users to understand the underlying assumptions and apply the models to their specific research questions. This curated library serves not only as a collection of ready-to-use tools but also as a valuable pedagogical resource, demonstrating best practices for constructing, fitting, and interpreting models, and providing robust templates for users aiming to develop novel model variants.

By providing a streamlined and efficient environment for the end-to-end Bayesian workflowfrom model specification and fitting to diagnostics and prediction, *BI* lowers the barrier to entry for sophisticated Bayesian modeling. We aim to empower a broader community of researchers across disciplines to confidently apply advanced Bayesian methods to their complex research problems; leveraging open-source, industry-standard frameworks such as NumPyro and TensorFlow Probability. Given this, our manuscript provides a substantial contribution to the field of statistical and a good resource for the community.

We have not submitted this manuscript for consideration at any other journal and a preprint version is available in bioRxiv. All computational code and data needed to reproduce our analyses and results will be made publicly available on an online repository. Given the focus of the manuscript, we believe that Matthew J. Silk (Matthew.Silk@ed.ac.uk), Jordan Hart (jordan.da.hart@gmail.com), Elizabeth A. Hobson (hobsoneh@ucmail.uc.edu) would make excellent potential reviewers.

In sum, we believe that *Methods in Ecology and Evolution* is the perfect outlet for our manuscript, as the resources we are providing with this suit of are key advances that will be relevant and impactful for the broad readership of the journal.

Sincerely,

Sebastian Sosa (On behalf of all coauthors), PhD

DATA AVAILABILITY STATEMENT

All implementations of *BI* are publicly available through their respective GitHub repositories for Python, R, and Julia. In addition, each implementation can be installed via the standard package repositories of the corresponding programming languages: pip for the Python version, CRAN for the R version, and the Julia package registry for the Julia version. Comprehensive documentation, including installation instructions, tutorials, and extended examples, are available on the official project website: https://s-sosa.com/BI.

AUTHORS' CONTRIBUTIONS

Sebastian Sosa conceived the research idea, developed the software, and co-authored the manuscript. Mary B. McElreath contributed to the writing of the manuscript. Cody T. Ross contributed to the development of the network models, the documentation of the models included in the software, and the writing of the manuscript.

FUNDING INFORMATION

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